

System Documentation

Study Title: Development and Use of a Simple Digital Tool for Assessing the Relationship Between Nursing Workload and Infection Prevention Compliance Among Nurses in Medical Wards of UNIOSUNTH

Version: 1.0 | **Date:** 2026

Table of Contents

- [1. System Overview](#)
- [2. Assessment Instruments](#)
- [3. Scoring Methods](#)
- [4. Statistical Analysis](#)
- [5. Data Flow & Architecture](#)
- [6. Multi-Device & Cloud Sync](#)
- [7. Security & Access Control](#)
- [8. PWA & Offline Capability](#)
- [9. File Structure](#)
- [10. Limitations & Recommendations](#)

1. System Overview

The UNIOSUNTH Nursing Research Tool is a **Progressive Web Application (PWA)** built for data collection and analysis in a cross-sectional nursing research study. It enables:

- **Nurses** to complete two validated self-report scales on their own mobile devices
- **The research coordinator** to aggregate, analyse, and export all submitted data from a central device

The tool operates **entirely offline** — assessments can be completed without internet access, and data syncs to the cloud database automatically when connectivity is restored.

Design Principles

Principle	Implementation
Anonymous data collection	Auto-generated nurse codes (no names stored)
Offline-first	All data stored locally; synced when online
Validated instruments	CSPS (Lam, 2004) + adapted NAS/NASA-TLX
Coordinator-only aggregation	PIN-protected cloud sync
Portable	Installable PWA — works on any smartphone or laptop

2. Assessment Instruments

2.1 Nursing Workload Scale (12 items)

Basis: Adapted from the Nursing Activities Score (Miranda et al., 2003) and the NASA Task Load Index (Hart & Staveland, 1988), contextualised for general medical wards in low- and middle-income country (LMIC) settings.

Response format: 5-point Likert scale

Value	Label
1	Not at all
2	Slightly
3	Moderately
4	Very much
5	Extremely

Subscales and items:

Subscale A — Physical & Task Demands (Items 1–4)

#	Item
1	Physical demands (lifting, moving, repositioning patients) were burdensome
2	Number of patients exceeded what I could comfortably manage
3	Had to perform complex or multiple procedures simultaneously or in rapid succession
4	Experienced time pressure — insufficient time to complete all required tasks

Subscale B — Cognitive & Emotional Demands (Items 5–8)

#	Item
5	Mental concentration required to monitor patients and make clinical decisions was high
6	Experienced emotional stress related to patient conditions or family interactions
7	Frequently interrupted or distracted while performing nursing tasks
8	Unexpected emergencies or deteriorating patients increased workload significantly

Subscale C — Administrative & Resource Burden (Items 9–12)

#	Item
9	Documentation, records, and administrative tasks consumed a considerable portion of shift
10	Inadequate supplies, equipment, or resources made nursing care harder
11	Coordination with other health team members added to workload
12	Overall workload was excessive relative to available nursing staff

No reverse-scored items in the workload scale. All items are positively keyed (higher = more burden).

2.2 IPC Compliance Scale — CSPS Adapted (20 items)

Basis: Compliance with Standard Precautions Scale (CSPS), originally developed by Dr. Simon Ching Lam (2004). The adapted version used here aligns with validations conducted in Nigerian and African hospital settings (Lam, 2014; Nigerian COVID-19 compliance studies, 2021–2023).

Response format: 4-point Likert scale

Value	Label
1	Never
2	Rarely
3	Sometimes
4	Always

Reverse-scored items: Items 3, 8, 10, and 11 describe non-compliant behaviours (e.g., recapping needles). For these items, scoring is inverted: **Never = 4, Always = 1**.

Subscales and items:

Subscale 1 — Personal Protective Equipment (Items 1–7)

#	Item	Reversed?
1	Wear gloves when at risk of contact with blood/body fluids	No
2	Change gloves and perform hand hygiene between patients	No
3	Reuse single-use gloves after cleaning/disinfecting	Yes
4	Wear gown/apron when at risk of splashing onto clothing	No
5	Wear surgical mask during splash-risk procedures	No
6	Wear eye protection during splash-risk procedures	No
7	Remove PPE appropriately and perform hand hygiene after	No

Subscale 2 — Sharps Safety (Items 8–12)

#	Item	Reversed?
8	Recap used needles with two-handed technique	Yes
9	Dispose of used sharps into puncture-resistant container immediately	No
10	Bend or break used needles before disposing	Yes
11	Pass used sharps directly from hand to hand	Yes
12	Report needlestick/sharps injuries promptly	No

Subscale 3 — Decontamination & Waste Management (Items 13–16)

#	Item	Reversed?
13	Clean blood/body fluid spills promptly with appropriate disinfectant	No
14	Ensure reusable equipment is decontaminated before use on another patient	No
15	Handle soiled linen without shaking; place in designated bag	No
16	Segregate and dispose of clinical waste in appropriate containers	No

Subscale 4 — Hand Hygiene & Cross-Infection Prevention (Items 17–20)

#	Item	Reversed?
17	Perform hand hygiene before direct patient contact	No
18	Perform hand hygiene after direct patient contact	No
19	Apply standard precautions for ALL patients regardless of diagnosis	No
20	Maintain aseptic technique during invasive procedures	No

3. Scoring Methods

3.1 Workload Score

Raw score range: 12 (minimum, all items rated 1) to 60 (maximum, all items rated 5)

Normalisation formula:

$$\begin{aligned}\text{Workload Score (\%)} &= ((\text{raw_total} - 12) / (60 - 12)) \times 100 \\ &= ((\text{raw_total} - 12) / 48) \times 100\end{aligned}$$

Subscale scores are computed the same way per subscale (each has 4 items, so min=4, max=20):

$$\begin{aligned}\text{Subscale Score (\%)} &= ((\text{subscale_raw} - 4) / (20 - 4)) \times 100 \\ &= ((\text{subscale_raw} - 4) / 16) \times 100\end{aligned}$$

Workload categories:

Score Range	Category	Clinical Implication
0–24%	Low	Manageable workload; IPC practices can be fully maintained
25–49%	Moderate	Moderate demand; occasional time pressure but IPC generally maintained
50–74%	High	High demand; risk that IPC shortcuts may be taken
75–100%	Very High	Critical overload; significant risk of IPC non-compliance

3.2 IPC Compliance Score (CSPS)

Scoring of reverse items: Items 3, 8, 10, 11 are inverted before summing:

Adjusted score = 5 - raw_score

(e.g., raw score of 4 "Always" for recapping needles becomes 1)

Raw score range: 20 (minimum) to 80 (maximum)

Normalisation formula:

IPC Score (%) = ((raw_total - 20) / (80 - 20)) × 100
= ((raw_total - 20) / 60) × 100

Subscale scores:

Each subscale has a different number of items (7, 5, 4, 4). The formula generalises as:

Subscale Score (%) = ((subscale_raw - n) / (n × 3)) × 100

where n = number of items in the subscale (min score = n × 1, max score = n × 4).

IPC compliance categories:

Score Range	Category	Interpretation
0–49%	Poor	Significant gaps in IPC practice; immediate training required
50–69%	Suboptimal	Inconsistent practice; targeted intervention needed
70–89%	Satisfactory	Generally good compliance with minor gaps
90–100%	Optimal	Consistently excellent compliance with all standard precautions

Thresholds adapted from: Lam (2014); WHO IPC assessment frameworks.

4. Statistical Analysis

4.1 Descriptive Statistics

For both workload and IPC scores across all participants, the tool computes:

Statistic	Formula
Mean	$\Sigma x_i / n$
Standard Deviation	$\sqrt{[\Sigma (x_i - \bar{x})^2 / (n-1)]}$ — sample SD
Median	Middle value of sorted scores (average of two middle values if n is even)
Minimum / Maximum	Lowest and highest scores in the dataset

4.2 Spearman's Rank Correlation Coefficient (ρ)

Why Spearman and not Pearson?

Spearman's ρ is used because:

1. The outcome variables (workload and IPC scores) are derived from Likert-scale responses, which are ordinal in nature
2. The distribution of scores may not be normally distributed (especially in small samples)
3. Spearman is robust to outliers — important in a hospital setting where extreme workload cases may occur
4. Consistent with the statistical approach used in comparable Nigerian nursing workload-IPC studies (e.g., Oyedele et al., 2023)

Computation:

Step 1 — Rank both variables independently (ties receive averaged ranks):

$R(x_i)$ = rank of workload score for participant i
 $R(y_i)$ = rank of IPC score for participant i

Step 2 — Compute sum of squared rank differences:

$$\sum d_i^2 = \sum [R(x_i) - R(y_i)]^2$$

Step 3 — Apply Spearman's formula:

$$\rho = 1 - (6 \times \sum d_i^2) / (n \times (n^2 - 1))$$

Range: -1.0 (perfect negative) to +1.0 (perfect positive). $\rho = 0$ means no monotonic relationship.

Expected hypothesis for this study:

- H_0 : There is no significant relationship between nursing workload and IPC compliance ($\rho = 0$)
- H_1 : There is a significant negative relationship ($\rho < 0$) — higher workload → lower compliance

4.3 Statistical Significance (p-value)

The p-value is computed using the **t-distribution approximation**:

$$t = \rho \times \sqrt{(n - 2) / (1 - \rho^2)}$$

with degrees of freedom = $n - 2$. The two-tailed p-value is derived from the incomplete beta function using the Wilson-Hilferty approximation (Numerical Recipes, Press et al.).

Significance threshold: $\alpha = 0.05$ (two-tailed)

Interpretation table (Cohen, 1988):

| $|\rho|$ range | Interpretation | |---|---| | < 0.10 | Negligible | | $0.10 - 0.29$ | Weak | | $0.30 - 0.49$ | Moderate | | $0.50 - 0.69$ | Strong | | ≥ 0.70 | Very strong |

Minimum sample size: A minimum of 3 records is required to display the correlation. For reliable inference in a research publication, a sample of **≥ 30 participants** is recommended (Bonett & Wright, 2000).

4.4 Visual Outputs

Chart	Type	Purpose
Scatter plot	XY scatter	Shows individual workload vs IPC pairs; dot colour = workload category

Category distribution	Grouped bar	Frequency of each workload/IPC category across participants
Subscale averages	Bar chart	Mean score for each of the 7 subscales
Records table	Table	Raw scores and categories for each participant

5. Data Flow & Architecture

5.1 Technology Stack

Layer	Technology	Purpose
Frontend framework	React 18 + TypeScript	Component-based UI with type safety
Build tool	Vite 5	Fast development server and optimised production build
Styling	Tailwind CSS 3	Utility-first responsive design
Routing	React Router v6	Client-side navigation between pages
Charts	Recharts 2	Responsive SVG-based data visualisations
Cloud database	Supabase (PostgreSQL)	Remote data storage and sync
Service worker	Workbox (via vite-plugin-pwa)	Offline caching and PWA functionality
Local storage	Browser localStorage	On-device data persistence
Cryptography	Web Crypto API	SHA-256 PIN hashing; random code generation

5.2 Application Pages

Route	Page	Description
/	Home	Overview, instrument summary, quick stats
/assess	Demographics	Auto-generated nurse code, ward, shift, qualification, patient load
/assess/workload	Workload Scale	12 Likert items grouped by subscale
/assess/ipc	IPC Scale	20 Likert items grouped by subscale
/assess/results	Results	Scores, categories, interpretation, save/submit actions
/data	Data Manager	All local records; cloud sync (coordinator only)
/analysis	Analysis	Descriptive stats, Spearman correlation, charts

5.3 Assessment Data Model

```
AssessmentRecord {
  id: string // Internal UUID (timestamp + random)
  timestamp: string // Human-readable date/time
  demographics: {
    nurseCode: string // Auto-generated e.g. "NRS-A7X2K9"
    ward: string
    shift: 'Morning' | 'Afternoon' | 'Night'
    qualification: 'RN' | 'BNSc' | 'RN+BNSc' | 'MSc' | 'PhD'
    yearsExperience: string
    patientLoad: string
  }
  workloadResponses: { [itemId]: 1|2|3|4|5 } // Raw responses
  ipcResponses: { [itemId]: 1|2|3|4 } // Raw responses
  workloadScore: number // 0-100 normalised
  ipcScore: number // 0-100 normalised
  workloadCategory: string // Low | Moderate | High | Very High
  ipcCategory: string // Poor | Suboptimal | Satisfactory | Optimal
  subscoreWorkload: { [subscale]: number } // 0-100 per subscale
  subscoreIPC: { [subscale]: number } // 0-100 per subscale
}
```

5.4 Supabase Database Schema

table: assessment_records

id	TEXT	PRIMARY KEY
submitted_at	TIMESTAMPTZ	DEFAULT now()
nurse_code	TEXT	NOT NULL
ward	TEXT	NOT NULL
shift	TEXT	NOT NULL
qualification	TEXT	NOT NULL
years_experience	TEXT	NOT NULL
patient_load	TEXT	NOT NULL
assessment_date	TEXT	NOT NULL
workload_score	INTEGER	NOT NULL
ipc_score	INTEGER	NOT NULL
workload_category	TEXT	NOT NULL
ipc_category	TEXT	NOT NULL
subscore_workload	JSONB	NOT NULL
subscore_ipc	JSONB	NOT NULL
workload_responses	JSONB	NOT NULL
ipc_responses	JSONB	NOT NULL

Row Level Security (RLS) policies:

- nurses_insert — anonymous users can INSERT (submit assessments)
- coordinator_select — anonymous users can SELECT (read all records)

6. Multi-Device & Cloud Sync

6.1 Nurse Workflow

1. Nurse installs PWA on their phone (or opens in browser)
 2. Opens app → "Start New Assessment"
 3. Auto-generated code (e.g. NRS-A7X2K9) assigned
 4. Completes workload scale (≈3 min) + IPC scale (≈4 min)
 5. Views individual results
 6. Taps "Submit to Study Database"
 - └ If online → uploads immediately → "Submitted ✓"
 - └ If offline → queued locally → auto-uploads when online

6.2 Coordinator Workflow

1. Opens app on any device
 2. Goes to Data tab
 3. Taps "Coordinator access" → enters PIN
 4. Taps "Sync from Cloud" → all nurses' records downloaded
 5. Opens Analysis tab → runs correlation analysis
 6. Exports CSV for SPSS/Excel
 - OR prints individual PDF reports per nurse

6.3 Offline Queue

Records that could not be submitted (no internet) are stored in `localStorage` under `uniosunth_upload_queue` . The app listens for the browser's `online` event and automatically flushes the queue when connectivity is restored. Each queued record shows a "Queued" badge in the Data Manager until uploaded.

6.4 CSV Export Columns

Column	Description
ID	Internal record identifier
Date	Assessment date/time
Nurse Code	Auto-generated anonymous code
Ward	Selected ward
Shift	Morning / Afternoon / Night
Qualification	Highest nursing qualification
Years Experience	Years in nursing
Patient Load	Number of patients that shift
Workload Score (%)	Normalised 0–100
Workload Category	Low / Moderate / High / Very High
IPC Score (%)	Normalised 0–100
IPC Category	Poor / Suboptimal / Satisfactory / Optimal

Workload subscores (3 columns)	Per subscale %
IPC subscores (4 columns)	Per subscale %

7. Security & Access Control

7.1 Coordinator PIN

The coordinator PIN protects cloud sync and aggregate data access. It is stored as a **SHA-256 hash** in the browser's localStorage — the plain PIN is never saved anywhere.

Process:

1. First use: coordinator sets a 4–8 digit PIN
2. PIN is hashed: `SHA-256(pin)` → stored in localStorage
3. Each unlock attempt: `SHA-256(entered_pin)` is compared to the stored hash
4. On match: session storage flag is set — unlocked for the current browser session
5. Closing the browser/tab automatically re-locks (session storage is cleared)

Reset: Delete the `uniosunth_coordinator_pin_hash` key from localStorage in browser DevTools.

7.2 Data Privacy

- No personally identifiable information (PII) is collected — only anonymous nurse codes
- Nurse codes are randomly generated (`crypto.getRandomValues`) and not linked to any identity
- Raw item responses are stored in Supabase for audit and re-analysis purposes
- Supabase project is configured with RLS — no unauthenticated DELETE or UPDATE operations are permitted

7.3 Nurse Code Generation

Codes follow the format `NRS-XXXXXX` where each `X` is drawn from a 32-character unambiguous alphabet:

```
ABCDEFGHIJKLMNOPQRSTUVWXYZ23456789
```

Characters `0`, `o`, `1`, `I` are excluded to prevent transcription errors. Using `crypto.getRandomValues` (cryptographically secure random), the probability of two identical codes being generated is:

```
P(collision) = 1 / 32^6 ≈ 1 in 1,073,741,824
```

This makes accidental duplication effectively impossible across a study sample.

8. PWA & Offline Capability

8.1 What is a PWA?

A Progressive Web App (PWA) is a website that can be installed on a device and behaves like a native app — it works offline, appears on the home screen, and runs in full-screen without a browser address bar.

8.2 Installation

Android (Chrome):

1. Open the app URL in Chrome

- 2. Tap the "Install" banner at the bottom, OR tap : → "Add to Home Screen"

iOS (Safari):

- 1. Open the app URL in Safari
- 2. Tap the Share button (□1) → "Add to Home Screen"

Desktop (Chrome/Edge):

- 1. Click the install icon (⊕) in the address bar

8.3 Offline Strategy

The service worker (generated by Workbox) pre-caches all application assets on first install:

Asset type	Strategy
HTML, JS, CSS	Cache-first (served from cache, updated in background)
Navigation fallback	Serves <code>index.html</code> for all routes — SPA routing works offline
Google Fonts	Cache-first with 1-year TTL (if used)

Data persistence offline: All assessment responses are saved to `localStorage` immediately on each interaction — if the browser crashes mid-assessment, no data is lost.

8.4 Updates

When a new version of the app is deployed:

- 1. The service worker detects the update in the background
- 2. An "Update Available" toast appears
- 3. Tapping "Update now" refreshes to the new version

9. File Structure

```
UNIOSUNTH-RESEARCH/
├─ index.html           # PWA entry point (meta tags, theme-color)
├─ vite.config.ts       # Build config + PWA plugin + chunk splitting
├─ tailwind.config.js   # Design system colours
├─ .env                 # Supabase credentials (not committed to git)
├─ .env.example         # Template for deployment
├─ SUPABASE_SETUP.sql   # SQL to run once in Supabase dashboard
├─
├─ public/
│  └─ favicon.svg       # App icon (heartbeat line on blue)
│     └─ apple-touch-icon.svg # iOS home screen icon
│
└─ src/
   └─ main.tsx          # App entry; registers auto-sync listener
   └─ App.tsx           # Router + PWA prompts
   └─ index.css         # Tailwind base + custom utilities
   └─
   └─ types/
```

```

├── index.ts                # AssessmentRecord, Demographics, etc.
├── data/
│   ├── workloadItems.ts    # 12 workload scale items + labels + subscales
│   └── ipcItems.ts         # 20 COPS items + labels + subscales + reversed flags
├── lib/
│   └── supabase.ts         # Supabase client (null if env vars not set)
├── utils/
│   ├── scoring.ts          # calcWorkloadScore, calcIPCScore, categories
│   ├── statistics.ts       # spearmanCorrelation, p-value, descriptives
│   ├── storage.ts          # localStorage CRUD + generateNurseCode
│   ├── sync.ts             # Upload, download, offline queue, auto-flush
│   ├── coordinator.ts      # PIN hash/verify, session lock/unlock
│   └── export.ts           # CSV export, printable HTML report
├── components/
│   ├── Layout.tsx          # Header, mobile bottom nav, coordinator banner
│   ├── ProgressBar.tsx     # Step indicator (dots desktop, label mobile)
│   ├── LikertItem.tsx      # Single scale question with response buttons
│   ├── ScoreCard.tsx       # Score display with category + subscore bars
│   ├── CoordinatorModal.tsx # PIN entry / setup bottom sheet
│   ├── PWAInstallPrompt.tsx # "Add to Home Screen" install banner
│   └── PWAUpdatePrompt.tsx  # "New version available" toast
├── pages/
│   ├── Home.tsx            # Landing page
│   ├── Demographics.tsx    # Step 1: auto code + ward/shift/qualification
│   ├── WorkloadAssessment.tsx # Step 2: 12 Likert items
│   ├── IPCAssessment.tsx   # Step 3: 20 Likert items
│   ├── Results.tsx         # Step 4: scores + submit button
│   ├── DataManager.tsx     # Records table/cards + coordinator sync
│   └── Analysis.tsx        # Stats + charts + records table

```

10. Limitations & Recommendations

10.1 Instrument Limitations

Limitation	Note
Self-report bias	Nurses self-report their IPC compliance — social desirability bias may inflate scores. Consider pairing with direct observational audits
Single-shift snapshot	The workload scale covers the most recent completed shift, not an average — scores may vary across shifts
Cross-sectional design	Cannot infer causation — only association between workload and compliance
Workload scale adaptation	The workload scale is adapted, not independently validated in a Nigerian medical ward context. Cronbach's alpha should be computed on the collected sample and reported

Sample size	Spearman's ρ is reliable from $n \geq 30$. For power to detect a moderate effect ($\rho = 0.30$) at $\alpha = 0.05$, two-tailed, 80% power, the required $n \approx 84$ participants
-------------	---

10.2 Technical Limitations

Limitation	Note
Browser storage limits	localStorage is limited to ~5–10 MB per origin — sufficient for thousands of records but not unlimited
No data backup	If a nurse clears their browser data before submitting, their record is lost
PIN security	The coordinator PIN is stored as a hash in localStorage — if a nurse has physical access to the coordinator's device, they could clear it and reset. For higher security, use Supabase Auth
Offline sync conflicts	If the same record ID is somehow submitted from two devices, the upsert operation will overwrite the earlier version

10.3 Recommended Reporting Template

When reporting results from this tool in a thesis or publication:

"Nursing workload was assessed using a 12-item scale adapted from the Nursing Activities Score (Miranda et al., 2003) and the NASA Task Load Index (Hart & Staveland, 1988) for general medical ward contexts. Infection prevention compliance was measured using the 20-item Compliance with Standard Precautions Scale (CSPS; Lam, 2004), previously validated in Nigerian hospital settings. Both scales were administered digitally via a purpose-built Progressive Web Application. Scores were normalised to 0–100% and categorised using established thresholds. The relationship between nursing workload and IPC compliance was examined using Spearman's rank correlation coefficient (ρ), with statistical significance set at $\alpha = 0.05$ (two-tailed). Data were analysed using built-in statistical functions within the tool, and exported to SPSS v.XX for verification."

References

- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum.
- Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX: Results of empirical and theoretical research. *Human Mental Workload*, 1, 139–183.
- Lam, S. C. (2004). *Compliance with Standard Precautions: Development of an Instrument*. (Doctoral dissertation). Hong Kong Polytechnic University.
- Lam, S. C. (2014). Validation and cross-cultural pilot testing of compliance with standard precautions scale. *Applied Nursing Research*, 27(3), 169–175.
- Miranda, D. R., Nap, R., de Rijk, A., Schaufeli, W., & Iapichino, G. (2003). Nursing activities score. *Critical Care Medicine*, 31(2), 374–382.
- Press, W. H., Teukolsky, S. A., Vetterling, W. T., & Flannery, B. P. (2007). *Numerical Recipes: The Art of Scientific Computing* (3rd ed.). Cambridge University Press.
- WHO (2009). *WHO Guidelines on Hand Hygiene in Health Care*. World Health Organization.